Mohammad Rahmanitalab

B.Sc. Mechanical Engineering, Sharif University of Technology
Minor in Computer Science, Sharif University of Technology
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Mohammad Rahmanitalab | MRahmaniT

Education

B.Sc. Major in Mechanical Engineering, SUT| Tehran, Iran *2021 – 2026*
Advisor: Prof. Saeed Behzadipour

- GPA: 3.95/4.00 (18.93/20) — Ranked **12th out of 116** students in the department.

B.Sc. Minor in Computer Science, SUT| Tehran, Iran *2022 – 2026*

- GPA: 4.00/4.00 (18.98/20)

Diploma in Mathematics and Physics| Tehran, Iran *2018 – 2021*

- GPA: 4.00/4.00 (19.88/20)

Research Interests

- **Machine Learning:** Time-series and sensor-data learning, supervised/deep learning, representation learning
- **Robotics & Control:** Medical and rehabilitation robotics, haptics, teleoperation, robot motion and control
- **Biomedical Motion Analysis:** IMU-based sensing, biomechanics, movement classification and clinical applications

Undergraduate Research

Machin Learning Based Motion Analysis | Sharif University of Technology 2024 – 2026

- Conducting motion analysis using IMU sensors to develop evidence-based therapeutic protocols for children with Duchenne Muscular Dystrophy (DMD), supervised by Prof. Behzadipour.
- Leading the full pipeline: data collection, preprocessing, feature extraction, and movement classification using machine learning models (MATLAB, Python).
- Investigating movement biomarkers to support clinical decision-making in pediatric rehabilitation.

Publications

An Activity-Specific Machine Learning Framework for Classifying Simulated NSAA Performance Levels Using Wearable IMUs | ICBME 2026

- M. Rahmanitalab, et al. Manuscript in preparation for submission to ICBME.

Professional Experience

Back-End Developer | Moj Gostaran Adak, Tehran Oct – Dec 2025

- Developed a Java backend integrated with the Traccar GPS platform for reliable real-time device communication and data handling.
- Implemented a customized Go backend and RESTful APIs for the company's internal frontend and locker-tracking system deployed in the Hamrah Aval project.

Internships

Robotics Engineering | Sina Robotics & Medical Innovators Co., Tehran Jun – Sep 2024

- Worked on surgical robotics, deriving kinematic solutions and simulating master–slave robotic components with emphasis on haptic teleoperation (MATLAB, MSC ADAMS, TwinCAT3).

Teaching Assistant

Fundamentals of Automatic Control Design | Sharif University of Technology Fall 2025

- Supported instruction, assessment, and student guidance in controller design, PID tuning, and stability analysis.

Automatic Control | Sharif University of Technology Spring 2025

- Supported instruction, assessment, and student guidance in control theory, feedback systems, and Lyapunov stability analysis.

Technical Projects

Programming

Advanced Programming (Java) Spring 2025
Developed an online multiplayer game using object-oriented design, multithreading, and socket-based network programming.

Fundamentals of Programming (C) Spring 2022
Implemented a Twitter-like social media application with user management and file-based data persistence.

Robotics and Control

Robotics Fall 2024
Modeled and simulated a multi-DOF robotic manipulator in MATLAB/Simulink, studying forward/inverse kinematics and dynamic behavior.

Fundamental Automatic Control Design Fall 2024
Designed and tuned a PID controller for a ball-and-beam system; analyzed transient response and stability margins in MATLAB/Simulink.

Automatic Control Spring 2024
Designed a PID controller for a rotating governor; performed full system performance and stability analysis.

Dynamics of Machines Spring 2024
Simulated a pushing mechanism in MATLAB/Simulink to analyze forces, motion, and dynamic response.

Skills

Programming: Python, C, Java, Go, TwinCAT3
Simulation Tools: MATLAB (Simulink), ROS2, Gazebo
Machine Learning: Time-series preprocessing, feature extraction, supervised classification
Robotics & Control: System modeling, PID control, simulation, sensor data processing
CAD: SolidWorks, MSC ADAMS
Languages: Persian (Native), English (Excellent)

Awards

High School | Tehran, Iran

- Ranked 187th in the National University Entrance Exam (Konkur). 2021

Undergraduate | Sharif University of Technology

- Ranked 85th in the National Master's Entrance Exam (Konkur). 2025
- Qualified for direct admission to M.Sc. due to academic excellence. 2025 & 2026

Selected Courses

Mechanical Engineering:

- Robotics
- Dynamics of Machinery
- Mechanisms Design
- Automatic Control
- Fund of Automatic Control Design
- Measurement and Control Systems
- Electrical Eng. I–II

Computer Science:

- Advanced Programming
- Dtra Transmission & Networks
- Computer Simulation
- Operating Systems
- Linear Algebra
- Probability
- Discrete Mathematics

Extra-Curricular Activities

CARE Student Society: Media Director; managed media content, production, and team operations.
ICERS 2024 & Dept. Website: Contributed to IT/media coordination and developed/maintained the Mechanical Engineering Department website (WordPress).